

JEEHO RYOO

Assistant Professor

Olsen College of Engineering and Science, Fairleigh Dickinson University

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778-726-4199

RESEARCH INTERESTS

- **Computer architecture:** novel execution and dataflow mechanisms that improve performance and scalability in computing platforms.
- **Memory Systems:** advanced refresh control and access scheduling techniques that mitigate latency, energy, and security vulnerabilities in DRAM.
- **Performance, energy efficiency, and hardware/software co-design:** cross-layer optimization frameworks that coordinate runtime decisions with microarchitectural control to maximize various efficiency metrics.
- **Emerging workload and system characterization:** empirical workload studies of LLMs, graph analytics, and data-centric pipelines to inform future system and memory hierarchy designs.
- **Education and learning technology:** AI-driven learning support systems that strengthen conceptual modeling, self-regulated learning, and feedback transparency in engineering courses.

ACADEMIC APPOINTMENTS

Assistant Professor

May 2024 – Present

Olsen College of Engineering and Science, Fairleigh Dickinson University, Vancouver, BC

Instructor

August 2021 – April 2024

British Columbia Institute of Technology, Vancouver, BC, Canada

EDUCATION

The University of Texas at Austin, College of Engineering, Austin, TX

May 2017

Doctor of Philosophy in Electrical and Computer Engineering (Advisor: Lizy K. John)

The University of Texas at Austin, College of Engineering, Austin, TX

May 2014

Master of Science in Electrical and Computer Engineering (Advisors: Lizy K. John)

Cornell University, College of Engineering, Ithaca, NY

May 2011

Bachelor of Science in Electrical and Computer Engineering (Advisors: David H. Albonese and Christopher Batten)

PUBLICATIONS

• Conference Proceedings

Year: 2026

Weidong Zhang, Yongchan Jung, Muhammad Ali Khaliq, Enkhzaya Chuluunbaatar, Byeong Kil Lee, and **Jeeho Ryoo**, "Lesion-DDPM: Lesion-Enhanced 3D Diffusion for MS MRI Synthesis" *In the Proceedings of IEEE International Symposium on Biomedical Imaging (ISBI)*, London, UK, April, 2026 (under review)

Jing Yi Huang, S. Q. Yap, Michael Pin-Chuan Lin, Daniel Chang, and **Jeeho Ryoo**, "Towards Procedural Transparency in the Use of Large Language Models for Qualitative Analysis: A Systematic Review in Educational Research," *In the Proceedings of the International Conference on Intelligent Tutoring Systems (ITS)*, Pafos, Cyprus, June, 2026.

Gerald Tembrevilla, Weidong Zhang, G. Michael Bowen, Michael Pin-Chuan Lin, Darryl Roy Montebon, and **Jeeho Ryoo**, "Where are the teachers? Rethinking AR/VR research in teacher education" *In the*

Proceedings of the Annual Conference and Exposition American Society for Engineering Education (ASEE), Charlotte, NC, USA, June, 2026

Michael Pin-Chuan Lin, Gurdeep Jhaji, Daniel Chang, Fan Lin, Eric Poitras, and **Jeeho Ryoo**, "Generative AI's Role in Computer Science Classrooms: A Rapid Mapping Review," *In the Proceedings of the Washington DC Conference on Education (WCE)*, Washington D.C., USA, April, 2026.

Michael Pin-Chuan Lin, Eric Poitras, Gian Tembrevilla, G. Michael Bowen, Daniel H. Chang, Nikki Lobczowski, Jing Yi Huang, Yu-Feng Lan, and **Jeeho Ryoo**, "Amplifier or Substitute? A Rapid Review of Generative AI in Higher Education Active Learning Tasks and Self-Regulated Learning," *In the Proceedings of the Society for Information Technology & Teacher Education International Conference (SITE)*, Philadelphia, PA, USA, March, 2026.

Jeeho Ryoo, Yongchan Jung, Weidong Zhang, Muhammad Ali Khaliq, and Byeong Kil Lee, "Performance Analysis and Optimization of 3D Generative Diffusion Models across GPU Architectures" *In the Proceedings of IEEE International Conference on Performance Engineering (ICPE)*, Florence, Italy, May, 2026

Michael P. C. Lin, Eric Poitras, Gerald Tembrevilla, G. Mike Bowen, Nikki Lobczowski, Daniel Chang, Jia-Yin Huang, Yu-Fang Lan, and **Jeeho Ryoo**, "Amplifier or Substitute? A Rapid Review of Generative AI in Higher Education Active Learning Tasks and Self-Regulated Learning," *In the Proceedings of the Society for Information Technology & Teacher Education International Conference (SITE)*, Philadelphia, PA, March, 2026.

Year: 2025

Xianjing Zhang, **Jeeho Ryoo**^{*}, and Sourav Mukherjee^{*}, "Detection of Body Focused Repetitive Behaviors Using Deep Learning" *In the Proceedings of 7th IEEE Computers, Communications and IT Applications Conference (ComComAP)*, Madrid, Spain, December, 2025. (^{*}equal contribution)

Ruihao Li, Shagnik Pal, Vineeth Narayan Pullu, Prasoon Sinha, **Jeeho Ryoo**, Lizy K. John, and Neeraja J. Yadwadkar. "Oneiros: KV Cache Optimization through Parameter Remapping for Multi-tenant LLM Serving" *In the Proceedings of ACM Symposium on Cloud Computing (SoCC)*, Virtual, November, 2025

Jeeho Ryoo^{*}, Michael Pin-Chuan Lin^{*}, Sahil Rai, Wenhao He, Seong Min Park, and Marco Ho. "WIP: Multi-Agent Artificial Intelligence Model to Enhance Self-Regulated Learning and Conceptual Understanding in Computer Science Education" *In the Proceedings of IEEE Frontiers in Education Conference (FIE)*, Nashville, TN, USA, November, 2025. (^{*}equal contribution)

Shagnik Pal, **Jeeho Ryoo**, and Lizy K. John. "The Curious Case of Global Stable Loads" *In the Proceedings of IEEE International Symposium on Workload Characterization (IISWC)*, Irvine, CA, October, 2025

Marco Ho, Carly Orr, Rebecca Jeon, Michael Pin-Chuan Lin, and **Jeeho Ryoo**. "AI Literacy Through a Project-Based Learning Course" *In the Proceedings of IEEE Smart World Congress (SWC)*, Calgary, AB, Canada, August, 2025.

Michael Pin-Chuan Lin, Fuhua Lin, Yu-Feng Lan and **Jeeho Ryoo**. "Perceiving Generative AI in Teacher Practice: a Design-Based Case Study in a Graduate Course" *In the Proceedings of IEEE Smart World Congress (SWC)*, Calgary, AB, Canada, August, 2025.

Muge Zhang, Muhammad Ali Khaliq, Byeong Kil Lee, and **Jeeho Ryoo**. "Synthetic Magnetic Resonance Imaging Generation for the Diagnosis of Alzheimer's Disease using Machine Learning" *In the Proceedings of 8th IEEE International Conference on Multimedia Information Processing and Retrieval (MIPR)*, San Jose, CA, USA, August, 2025.

Michael Pin-Chuan Lin, Arita Li Liu, Saeed Saffari, Daniel Chang, and **Jeeho Ryoo**. "Mapping AI Tools in Education: A Topic Modeling Analysis of Cognitive, Metacognitive, & Affective Insights" *In*

Proceedings of the 21st International Conference on Intelligent Tutoring Systems (ITS), Alexandroupolis, Greece, June 2025

Seong Min Park, Marco Ho, Michael Pin-Chuan Lin, and **Jeeho Ryoo**. “Evaluating the Impact of Assistive AI Tools on Learning Outcomes and Ethical Considerations in Programming Education” *In the Proceedings of the 16th IEEE Global Engineering Education Conference (EDUCON)*, London, UK, April 2025.

Saeed Saffari, Michael Pin-Chuan Lin, Oscar Lin, and **Jeeho Ryoo**. “Investigating Open-Source Large Language Models in Digital Pedagogies” *In the Proceedings of the 18th International Conference on e-Learning & Innovative Pedagogies*, Changhua City, Taiwan, April 2025.

Year: 2024

Muge Zhang, Dae Yeol Lee, Vasudevan Janarthanan, and **Jeeho Ryoo**, “Microarchitectural Analysis of Pre-Processing Stage in Machine Learning Workloads” *In Proceedings of the 7th IEEE International Conference on Algorithms, Computing and Artificial Intelligence (ACAI)*, Guangzhou, China, December 2024.

Year: 2023

Kumar, Shvetha S., Reshma R. Nayak, Jismi S. Kannampuzha, Sahil Rai, **Jeeho Ryoo**, and Lizy K. John. "Evaluation of Pruning Techniques" *In the proceedings of the 43rd IEEE International Performance, Computing, and Communications Conference (IPCCC)*, Anaheim, CA, USA, November 2023.

Jensen, Steffen, Jaekyu Lee, Dam Sunwoo, Matt Horsnell, Matthew Siggs, **Jeeho Ryoo**, and Lizy John. "Do Video Encoding Workloads Stress the Microarchitecture?" *In the proceedings of the IEEE International Symposium on Workload Characterization (IISWC)*, Ghent, Belgium, October 2023.

Year: 2018

Jee Ho Ryoo, Shuang Song and Lizy K. John, “Puzzle Memory: A Multifractional Partitioned Heterogeneous Memory Scheme” *In the Proceedings of the 36th IEEE International Conference on Computer Design (ICCD)*, Orlando, FL, USA, October 2018.

Jee Ho Ryoo, Lizy K. John, and Arkaprava Basu, “A Case for Granularity Aware Page Migration” *In the Proceedings of the 32nd ACM International Conference on Supercomputing (ICS)*, Beijing, China, June 2018.

Year: 2017

Yashwant, Marathe, Nagendra Gulur, **Jee Ho Ryoo**, Shuang Song, and Lizy K. John, “CSALT: Context Switch Aware Large TLB” *In the Proceedings of the 50th ACM/IEEE International Symposium on Microarchitecture (MICRO)*, Boston, MA, USA, October 2017.

Jee Ho Ryoo, Nagendra Gulur, Shuang Song and Lizy K. John, “Rethinking TLB Designs in Virtualized Environments: A Very Large Part-of-Memory TLB” *In the Proceedings of the 44th ACM/IEEE International Symposium on Computer Architecture (ISCA)*, Toronto, ON, Canada, June 2017.

Jee Ho Ryoo, Mitesh R. Meswani, and Lizy K. John, “SILC-FM: Subblocked InterLeaved Cache-Like Flat Memory Organization” *In the Proceedings of the 23rd IEEE International Conference on High Performance Computer Architecture (HPCA)*, Austin, TX, USA, February 2017.

Year: 2016

Shuang Song, Meng Li, Xinnian Zheng, **Jee Ho Ryoo**, Reena Panda, Michael LeBeane, Andreas Gerstlauer, and Lizy K. John, “Proxy-Guided Load Balancing of Graph Processing Workloads on Heterogeneous Clusters” *In the Proceedings of the 45th International Conference on Parallel Processing (ICPP)*, Philadelphia, PA, USA, August 2016.

Reena Panda, Xinnian Zheng, **Jee Ho Ryoo**, Michael LeBeane, Shuang Song, Andreas Gerstlauer, and Lizy K. John, “Genesys: Automatically Generating Representative Training-sets” *In the Proceedings of*

the 16th International Conference on Embedded Computer Systems: Architecture, Modeling, and Simulation (SAMOS), Samos Island, Greece, July 2016.

Year: 2015

Michael LeBeane, Shuang Song, Reena Panda, **Jee Ho Ryoo**, and Lizy K. John, "Data Partitioning Strategies for Graph Workloads in Heterogeneous Clusters" *In the Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC)*, Austin, TX, USA, November 2015.

Jee Ho Ryoo, Karthik Ganesan, Yao-Min Chen, and Lizy K. John, "i-MIRROR: A Software Managed Die-Stacked DRAM-Based Memory Subsystem" *In the Proceedings of the 27th IEEE International Symposium on Computer Architecture and High Performance Computing (SBAC-PAD)*, Florianopolis, Brazil, October 2015.

Michael LeBeane, **Jee Ho Ryoo**, Reena Panda, and Lizy K. John, "WattWatcher: Fine-Grained Power Estimation For Emerging Workloads" *In the Proceedings of the 27th IEEE International Symposium on Computer Architecture and High Performance Computing (SBAC-PAD)*, Florianopolis, Brazil, October 2015.

Reena Panda, Christopher Erb, Michael LeBeane, **Jee Ho Ryoo**, and Lizy K. John, "Performance Characterization of Modern Databases on Out-of-order CPUs" *In the Proceedings of the 27th IEEE International Symposium on Computer Architecture and High Performance Computing (SBAC-PAD)*, Florianopolis, Brazil, October 2015.

Jee Ho Ryoo, Saddam Quireem, Michael LeBeane, Reena Panda, Shuang Song, and Lizy K. John, "GPGPU Benchmark Suites: How Well Do They Sample the Performance Spectrum" *In the Proceedings of the 44th International Symposium on Parallel Processing (ICPP)*, Beijing, China, September 2015. **(Best Paper Runner-Up)**

Wooseok Lee, Youngchun Kim, **Jee Ho Ryoo**, Dam Sunwoo, Andreas Gerstlauer, and Lizy K. John, "PowerTrain: A Learning-based Calibration of McPAT Power Models" *In the Proceedings of the ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED)*, Rome, Italy, July 2015.

Year: 2013

Mohammad Faisal Iqbal, Jim Holt, **Jee Ho Ryoo**, Gustavo De Veciance, and Lizy K. John, "Flow Migration on Multicore Network Processors: Load Balancing While Minimizing Packet Reordering" *In the Proceedings of the 42nd International Conference on Parallel Processing (ICPP)*, Lyon, France, October 2013.

Year: 2012

Jinsuk Chung, Ikhwan Lee, Michael Sullivan, **Jee Ho Ryoo**, Don Wan Kim, Doe Hyun Yoon, Larry Kaplan, and Mattan Erez, "Containment Domains: A Scalable, Efficient, and Flexible Resilience Scheme for Exascale Systems" *In the Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC)*, Salt Lake City, UT, USA, November 2012.

● **Journals**

Year: 2026

Michael Pin-Chuan Lin, Jing Yi Huang, Daniel H. Chang, Gian Tembrevilla, G. Michael Bowen, Eric Poitras, Vikram Janarthanan, and **Jeeho Ryoo**, "Mapping Open-Source Large Language Models in Education: A Narrative Review of Evidence, Pedagogical Roles, and Learning Outcomes," *AI in Education*, 2026.

Michael Pin-Chuan Lin, Daniel Chang, Vivien Lin, **Jeeho Ryoo**, "Meta-Analysis of Argument Visualization Tools in Higher Education: Examining Effects on Student Achievement and Moderating Factors" *In Technology, Knowledge and Learning*, 2026.

Year 2016

Mohammad Faisal Iqbal, Jim Holt, **Jee Ho Ryoo**, Gustavo de Veciana and Lizy K. John, "Dynamic Core Allocation and Packet Scheduling in Multicore Network Processors" *In IEEE Transactions on Computers*, vol. 65, no. 12, December, 2016.

Year: 2013

Jinsuk Chung, Ikhwan Lee, Michael Sullivan, **Jee Ho Ryoo**, Don Wan Kim, Doe Hyun Yoon, Larry Karplan and Mattan Erez, "Containment Domains: A Scalable, Efficient, and Flexible Resilience Scheme for Exascale Systems" *In the Journal of Scientific Programming*, vol. 21, no. 3-4, 2013.

● Workshop and Posters

Year: 2025

Cory Davis, Patrick Stockton, Eugene B. John, **Jeeho Ryoo**, and Ebod Shojaei, "Hardware Acceleration for Graph Neural Networks," In the Proceedings of the 23rd ACM International Conference on Computing Frontiers (Posters), 2026.

Year: 2025

Patrick Stockton, Cory Davis, **Jeeho Ryoo**, and Eugene John, "Accelerating Neuro-Symbolic AI with CUTLASS-Optimized CUDA Kernels" *8th IBM/IEEE AI Compute Symposium*, Yorktown Heights, NY, USA, November, 2025. (Poster)

Ebod Shojaei, Wenyun Dai, Anshul Jha, Eugene John, and **Jeeho Ryoo**, "Hardware-Based Reinforcement Learning for Adaptive Dynamic Voltage and Frequency Scaling" *8th IBM/IEEE AI Compute Symposium*, Yorktown Heights, NY, USA, November, 2025. (Poster)

Cory Davis, Patrick Stockton, **Jeeho Ryoo**, and Eugene John. "Improving Energy Efficiency of Graph Neural Network Execution by using a PIM Architecture" *The 1st International Workshop on Data Centers Energy Efficiency (DCEE-2025)*, Tokyo, Japan, June, 2025. (Workshop)

Cory Davis, Patrick M. Stockton, Muge Zhang, **Jeeho Ryoo**, and Eugene John, "Microarchitectural Characterization of LightGCN and ExpressGNN and Architectural Implications" *High Performance, Edge And Cloud computing (HiPEAC)*, Barcelona, Spain, January, 2025 (Poster)

Year: 2016

Jee Ho Ryoo, Mitesh R. Meswani, Reena Panda, and Lizy K. John, "SILC-FM: Subblocked InterLeaved Cache-Like Flat Memory Organization" *International Conference on Parallel Architecture and Compilation Techniques (PACT)*, Haifa, Israel, September, 2016.

Year: 2014

Jee Ho Ryoo, Michael LeBeane, Mohammad Faisal Iqbal, and Lizy K. John, "Control Flow Behavior of Cloud Workloads" *International Conference on Workload Characterization (IISWC)*, Raleigh, NC, USA October, 2014.

GRANTS AND IN-KIND CONTRIBUTIONS

Social Sciences and Humanities Research Council (SSHRC) – Insight Development Grants (IDG) (Under Review)

- Title: Emotion-Aware Learning Supports for Equitable Computing Education
- 99,040 CAD (2026-2028)
- Applicant: Michael Marco Ho (30%)
- Co-Applicant: Nikki Lobczowski (20%), Michael Pin-Chuan Lin (20%), Daniel Chang (20%)
- Collaborator: **Jeeho Ryoo** (10%)

National Science Foundation (NSF) – Future Computing Research (Future CoRe)(Under Review)

- Title: Collaborative Research: SHF: Architecture-Aware Hardware–Software Co-Design for Efficient Neuro-Symbolic AI Acceleration
- 1,000,000 USD (2026-2030)

- Principal Investigator: Eugene John (66%), **Jeeho Ryoo** (35%)

National Science Foundation (NSF) – Collaborative Research in Computational Neuroscience (CRCNS)(Under Review)

- Title: Hardware-Optimized Dual-Conditioned Diffusion Models Unifying Multicontrast MRI Synthesis and AD-MS Cross-Domain Learning
- 657,695 USD (2026-2029)
- Principal Investigator: Byeong Kil Lee (61%), **Jeeho Ryoo** (39%)

National Science Foundation (NSF) – Research Initiation in Engineering Formation (PFE: RIEF)(Under Review)

- Title: Hybrid Professional and Ethical Formation of Civil and Computer Engineers in Data-Driven Building and Infrastructure Inspection
- 200,000 USD (2026-2027)
- Principal Investigator: Hyungjoo Choi (13%), Husam Najm (21%), Eugene John (21%), Alexandra Patzak (35%)
- Co-Principal Investigator: **Jeeho Ryoo** (10%)

Social Sciences and Humanities Research Council (SSHRC) – Insight Grants (IG)

- Title: Tracing Self-Regulated Learning in Computer Science Education with an AI-Enhanced Learning Tool
- 415,539 CAD (2026-2030)
- Applicant: Michael Pin-Chuan Lin (20%)
- Co-Applicant: Maiga Chang (20%), Daniel Chang (20%)
- Collaborator: **Jeeho Ryoo** (20%), Marco Ho (10%), Philip Winne (10%)

Fairleigh Dickinson University Grant-in-Aid

- Title: Reinforcement Learning Approaches for Memory Vulnerability
- 2,000 USD (2025-2026)
- Principal Investigator: **Jeeho Ryoo** (100%)

Google Cloud Research Credits

- Title: Synthetic Input Generation for Alzheimer's Disease
- 6921.75 USD (2024-2025)
- Principal Investigator: **Jeeho Ryoo** (100%)

PROFESSIONAL EXPERIENCES

Senior Performance Engineer

February 2018 – March 2019

ARM Inc., San Jose, CA

- Implemented DRAM cache simulation infrastructure modeling future ARM server systems
- Developed caching and prediction schemes tailored for emerging memory technology

Senior Software Engineer

July 2017 – February 2018

Oracle Corporation, Santa Clara, CA

- Identified bottleneck in Transparent Huge Pages (THP) when running enterprise applications
- Analyzed cloud application performance across various hardware platforms

Research Engineering Intern

May 2016 – December 2016

Advanced Micro Devices, Austin, TX

- Proposed granularity aware migration schemes in heterogeneous memory systems
- Implemented dynamic granularity migration schemes on Linux kernel running on real x86-64 systems

Research Engineering Intern

September 2015 – December 2015

Advanced Micro Devices, Austin, TX

- Proposed bandwidth efficient and high capacity utilization die-stacked DRAM management policy
- Developed heterogeneous memory simulation platform incorporating emerging memory technologies

Software Engineering Intern

May 2013 – August 2013

Oracle Corporation, Santa Clara, CA

- Proposed hardware-assisted, software-managed die-stacked DRAM management policy
- Evaluated the proposed scheme with commercial benchmarks such as SPECjbb, SPECjEnterprise, TPC-C

Performance Architecture Intern

May 2012 – August 2012

Samsung Austin R&D Center, Austin, TX

- Developed the SimPoint tool for ARM ISA to analyze various Android benchmarks
- Evaluated various power-efficient floating point schedulers using the Gem5 simulator

Graduate Research Assistant

August 2011 – May 2017

Laboratory for Computer Architecture, The University of Texas at Austin, Austin, TX

- Proposed management policies for high bandwidth, low latency die-stacked DRAM technology
- Developed other uses of die-stacked DRAM such as a large TLB in virtual platforms

Undergraduate Research Assistant

June 2010 – May 2011

Computer Systems Laboratory, Cornell University, Ithaca, NY

- Developed a digital ASIC tool-flow chain to synthesize the Verilog MIPS R10K microprocessor model
- Incorporated clock gating techniques at the functional block level (e.g. ALU, ROB) for power efficiency

Project Team Co-Director – Big Red Chip

September 2010 – May 2011

Computer Systems Laboratory, Cornell University, Ithaca, NY

- Designed a superscalar OOO processor and on-board I/O interfaces on Xilinx Virtex-6 FPGA
- Provided project design guidelines for 5 sub-design teams and managed the project deadline

TEACHING EXPERIENCES

Assistant Professor in Olsen College of Engineering and Science

Fairleigh Dickinson University, Vancouver, BC

- CSCI 6806 – Computer Science Graduate Capstone Project (Fall 2025, Summer 2025, Spring 2025, Fall 2024, Summer 2024)
- CSCI 5565 – Assembly Programming (Fall 2025, Summer 2024)
- INFO 4201 – Information Technology Needs Assessment and Management (Fall 2025)

Faculty in the School of Computing and Academic Studies

British Columbia Institute of Technology, Vancouver, BC

- COMP 2510 – Procedural Programming in C (Winter 2022, Fall 2022, Winter 2023, Fall 2023)
- COMP 3522 – Object Oriented Programming 2 (Fall 2021, Winter 2022)
- COMP 4736 – Introduction to Operating Systems (Winter 2023)
- COMP 4800 – Projects Practicum 2 (Spring 2023)
- COMP 7035 – Operating Systems (Fall 2023)

Teaching Assistant

Electrical and Computer Engineering, The University of Texas at Austin, Austin, TX

- EE306 (Instructor: Yale Patt) – Introduction to Computing (Fall 2011)
- EE382N (Instructor: Lizy John) – Performance Evaluation and Benchmarking (Fall 2014)
- EE460N (Instructor: Mattan Erez) – Computer Architecture (Spring 2012)

Teaching Assistant

Electrical and Computer Engineering, Cornell University, Ithaca, NY

- ENGRD2300 (Instructors: David H. Albonesi, Adam W. Bojanczyk, José F. Martinez, and Edward G. Suh)
– Introduction to Digital Logic Design (Spring 2010, Fall 2010, Spring 2011)

ACADEMIC SERVICES AND PROFESSIONAL MEMBERSHIP

Professional Membership

- Member, IEEE
- Member, IEEE Education Society

- Member, ACM

Chair

- IEEE Frontier in Education (FIE) Session Ethics and Policy in AI Education: Developing Competencies for Responsible Technology Use: 2025 (Session Chair)
- IEEE International Workshop on Smart Education in the Age of Generative AI (SEGA): 2025 (General Co-Chair)

Technical Program Committee

- IEEE International Conference on High Performance Computer Architecture (HPCA): 2025
- IEEE International Conference on Performance, Computing and Communications (IPCCC): 2025, 2024
- IEEE International Conference on High Performance Computing, Data, and Analytics (HiPC'25): 2025
- IEEE International Parallel and Distributed Processing Symposium (IPDPS): 2025
- Supercomputing India (SCI): 2025

Reviewer

- ACM Computing Surveys: 2026, 2025, 2024
- ACM Transactions on Software Engineering and Methodology (TOSEM): 2025
- ACM/SIGAPP Symposium on Applied Computing (SAC): 2025
- IEEE Frontier in Education (FIE): 2026, 2025
- IEEE International Conference on High Performance Computing, Data, and Analytics (HiPC): 2025
- IEEE Transactions on Computers (TC): 2015
- ASEE Computers in Education (CoED): 2026
- Simulation Modelling Practice and Theory (SIMPAT): 2016

Student Volunteer

- IEEE International Symposium on Workload Characterization (IISWC): 2011
- ACM/SPEC International Conference on Performance Engineering (ICPE): 2015

UNIVERSITY SERVICES

Chair

- Research Release Time (RRT) Committee, Gildart Haase School of Computer Sciences and Engineering: 2025

Committee Member

- Office of Grants and Sponsored Project Research Council: 2025-Present
- Elections Committee, Olsen College of Engineering and Science: 2025-Present
- Search Committee for the Chair of the Department of Computer Science and Math, Olsen College of Engineering and Science: 2025
- Search Committee for the Director of Gildart Haase School of Engineering, Olsen College of Engineering and Science: 2025

Supervisor of Industry Sponsored Projects

- BinanceUS: Fall 2024
- UnifiedAI: Fall 2024

University Recruitment Representative

- FDU Vancouver Open House: Fall 2025
- QS Discover Master's Fair: Fall 2024

STUDENTS

Ph.D. Students

- Cory Davis (UTSA): 2024-Present
- Patrick Stockton (UTSA): 2024-Present
- Shafayat Mowla Anik (UCCS): 2025-Present

- Jing-Yuan (Alex) Huang (SFU): 2025-Present

M.S. Students

- Yongchan Jung (FDU): 2025-Present
- Dazhi (Alex) Yang (FDU): 2025-Present
- Seong Min Park, MS (FDU): 2025-Present

Past Students

- Jiatong Han (FDU): 2025-2026
- Xianjing (Chloe) Zhang (FDU): 2025
- Weidong Zhang, MS (Northeastern University): 2025 – 2026
- Ebod Shojaei (BCIT): 2024-2026
- Muhammad Ali Khaliq (UCCS): 2025-2025
- Sahil Rai, BS (BCIT): 2023-2025
- Jinxin Yin, MS (FDU): 2025
- Muge Zhang, MS (FDU): 2024-2025
- Rebecca Jeon, BS (University of Victoria): 2025

PATENTS

Intelligently Partitioning Data Cache to Allocate Space for Translation Entries

Inventors: Lizy K. John, Yashwant Marathe, **Jee Ho Ryoo**, and Nagendra Gulur.

US Patent US10261915B2, Issued on April 16, 2019

Methodology to Utilize Heterogeneous Memories with Variable Properties

Inventors: Lizy K. John, **Jee Ho Ryoo**, Hung-Ming Hsu, and Karthik Ganesan.

US Patent US20180260323A1, Issued on January 26, 2021

Page-Migration with Varying Granularity

Inventors: Arkaprava Basu and **Jee Ho Ryoo**.

US Patent US10503658B2, Issued on December 10, 2019

Processor Using a Level 3 Translation Lookaside Buffer Implemented in Off-Chip or Die-Stacked Dynamic Random-Access Memory

Inventors: Lizy K. John, **Jee Ho Ryoo**, and Nagendra Gulur

US Patent US10296465B2, Issued on May 21, 2019

Data Block Sizing for Channels in a Multi-Channel High-Bandwidth Memory

Inventors: **Jee Ho Ryoo** and Mitesh R. Meswani.

US Patent US10503655B2, Issued on December 10, 2019

Low Latency, High Bandwidth Memory Subsystem Incorporating Die-Stacked DRAM

Inventors: **Jee Ho Ryoo**, Karthik Ganesan, and Yao-Min Chen.

US Patent US20150279436A1, Issued on August 2, 2016

INVITED TALKS

“Future of Computing and AI Impact,” *Vancouver Korean IT Conference*, Vancouver, BC, July 15, 2023

AWARDS

Samsung Award of Excellence (2012) – Intern award recipient for excellent work performance

Engineering Learning Initiatives (2010) – Motorola research funding recipient for two semesters

Roger Berman '70 Memorial Prize (2010) – Top presentation student in engineering communications program

REFERENCES

Lizy K. John, Ph.D.

Professor, Truchard Foundation Chair in Engineering

Department of Electrical and Computer Engineering
University of Texas at Austin
EER 5.876, 2501 Speedway
Austin, TX 78712
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Michael S. Hsiao, Ph.D.

Professor
Department of Electrical and Computer Engineering
Virginia Tech
355 Durham, 1145 Perry Street
Blacksburg, VA 24061
Phone: 540-231-9254
Email: mhsiao@vt.edu

Byeong Kil Lee, Ph.D.

Associate Professor
Department of Electrical and Computer Engineering
University of Colorado – Colorado Springs
AEC 304, 1420 Austin Bluffs Pkwy
Colorado Springs, CO 80918
Phone: 719-255-4565
Email: blee@uccs.edu